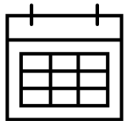
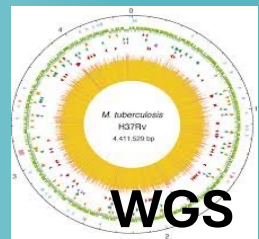
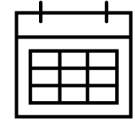


Molecular approaches: TB practical day

Monday 15 September 2025



MOLECULAR ASSAYS FOR TUBERCULOSIS: increased levels of complexity

INFORMATION	<i>Mtb</i> /no <i>Mtb</i> RIF	<i>Mtb</i> /no <i>Mtb</i> 15 drugs Genotype	<i>Mtb</i> /no <i>Mtb</i> all drugs Genotype Transmission
TARGETS	MTB-specific 1 (4) drug-target(s)	MTB-specific 18 drug-targets	4.4Mb All drug-targets
SAMPLE TYPE	Raw specimen	Treated Specimen ***	Cultured isolate
MOLECULAR ASSAY	Simple NAATs  	DEEPLEX tNGS  	WGS  

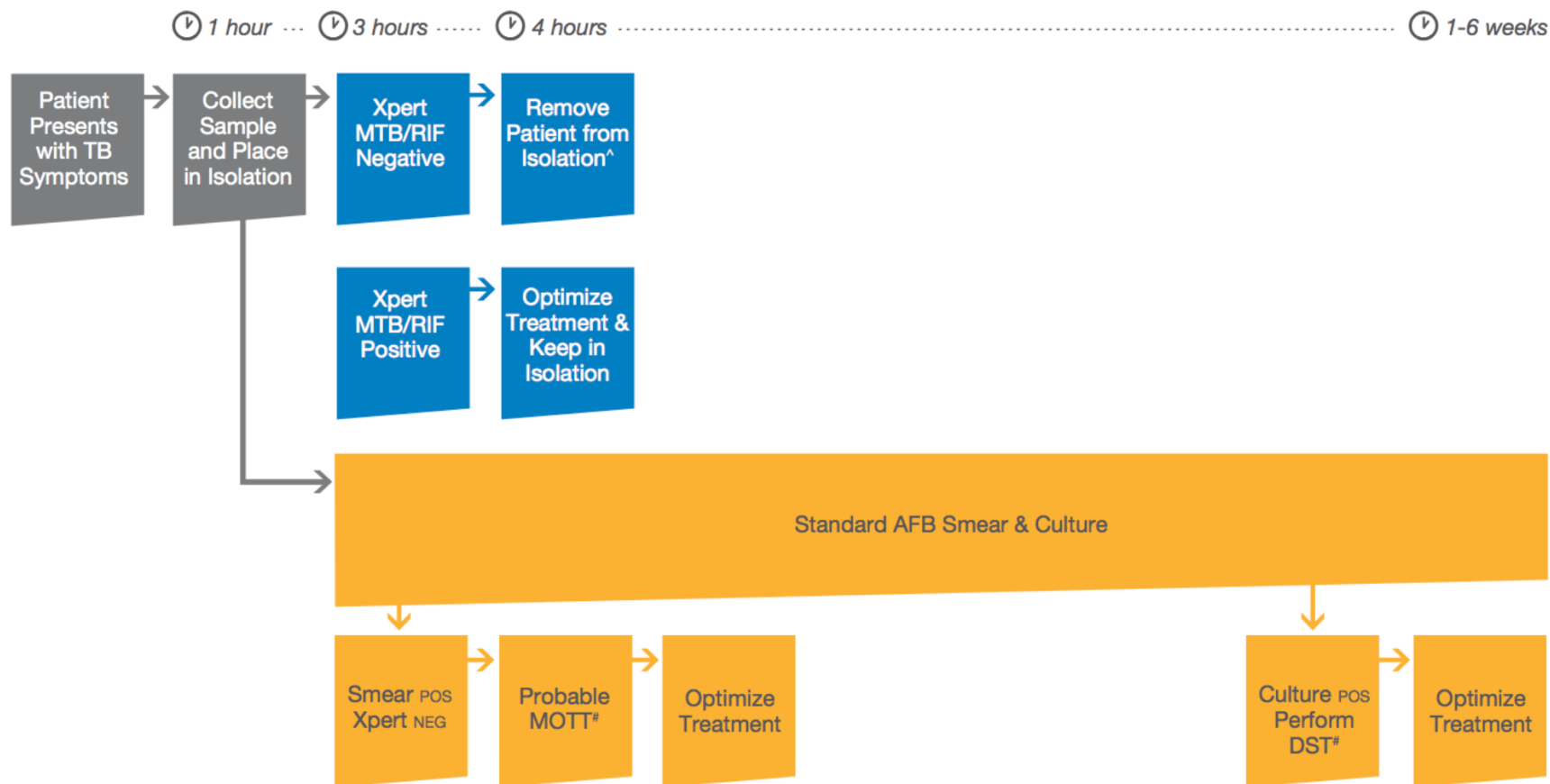
Cepheid GeneXpert® System

Nucleic acid amplification tests (NAATs)



The impact of Xpert on TB diagnosis and care

Impact on Patient Isolation Pathway
"Test and Release" or "Test and Treat"



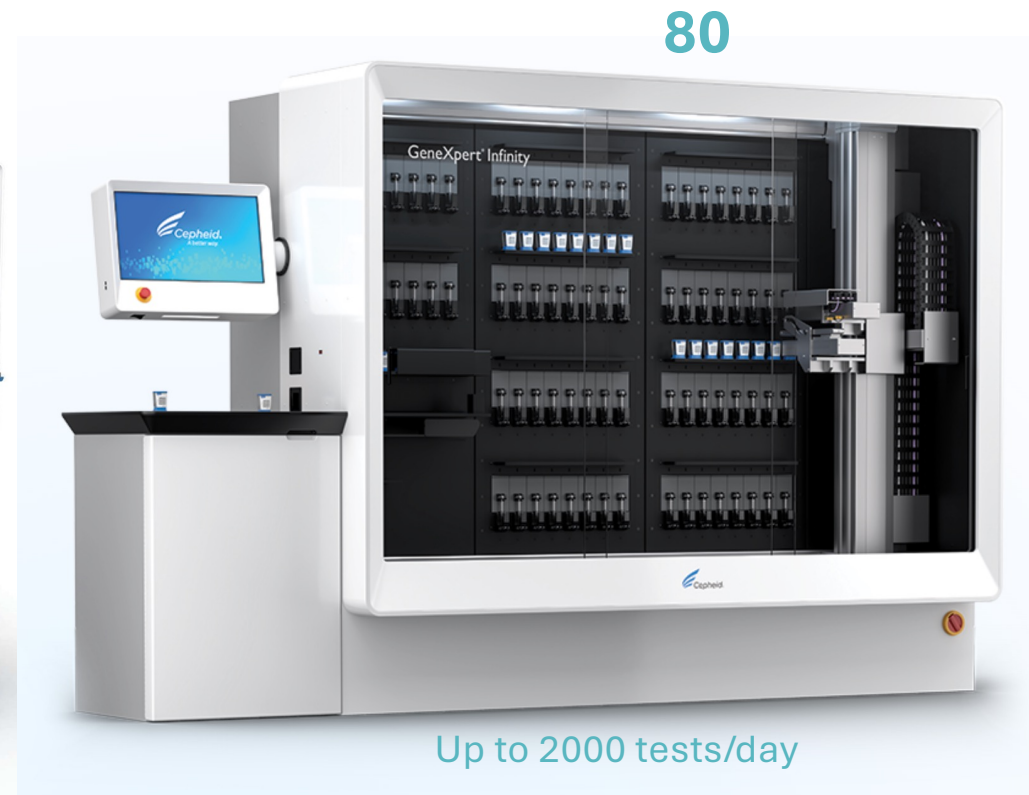
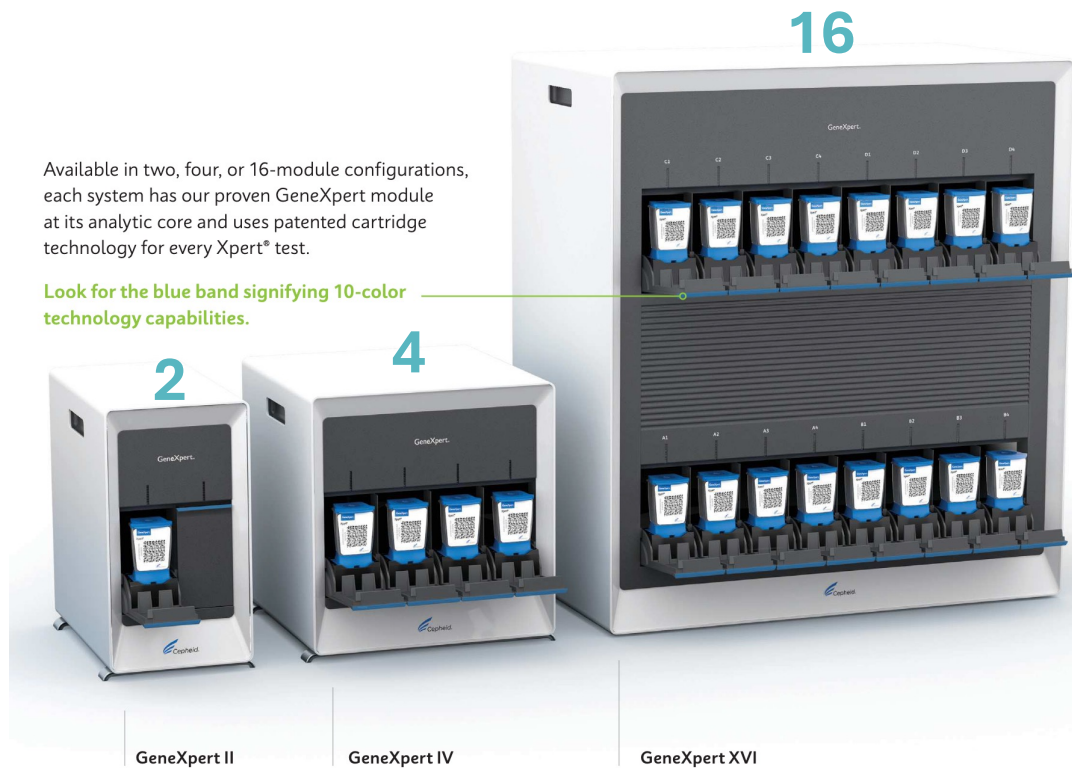
System vs. Assays - semantics

- THE SYSTEM: GeneXpert® System
 - “The system is designed to purify, concentrate, detect and identify targeted nucleic acid sequences, thereby delivering answers directly from unprocessed samples.”
- THE ASSAY: Xpert <<your favourite assay>>
 - Xpert® MTB/RIF
 - Xpert® MTB/XDR
 - Xpert® Xpress Flu
 - Xpert® Xpress Strep A

GeneXpert systems for different throughput needs

Available in two, four, or 16-module configurations, each system has our proven GeneXpert module at its analytic core and uses patented cartridge technology for every Xpert® test.

Look for the blue band signifying 10-color technology capabilities.



GeneXpert Infinity

How does it work?



▶ Journey Inside the Cepheid GeneXpert® Cartridge — 3D Animation

IVD In Vitro Diagnostic Medical Device. May not be available in all countries.



How does the PCR detection used by Xpert MTB/RIF Ultra work?

- TB or no TB? (That is the question)
 - Targets:
- Xpert MTB/RIF:
 - RIF susceptible or RIF resistant?
 - Target: *rpoB* – RNA polymerase
 - Rifampicin resistance determining region – 82 bp genomic region in *rpoB*
 - Wild type and mutant probes
- Xpert MTB/XDR
 - 10 colour-assay
 - Extended drug resistance:
 - INH
 - FQs
 - SLIDs
 - ETH

Xpert assays for TB



2006

Cepheid, FIND, Rutgers University, and NIAID begin development of **Xpert MTB/RIF** with funding from the Bill & Melinda Gates Foundation



2017

Xpert MTB/RIF Ultra launches and gets WHO endorsement



2020

Xpert MTB/XDR launched on new 10-color multiplexing technology

2021

Xpert MTB/XDR gets WHO endorsement

Interpreting results

Sample Type: Pericardial Fluid

Assay Information

Assay	Assay Version	Assay Type
Xpert MTB-RIF Assay G4	5	In Vitro Diagnostic

Test Result:

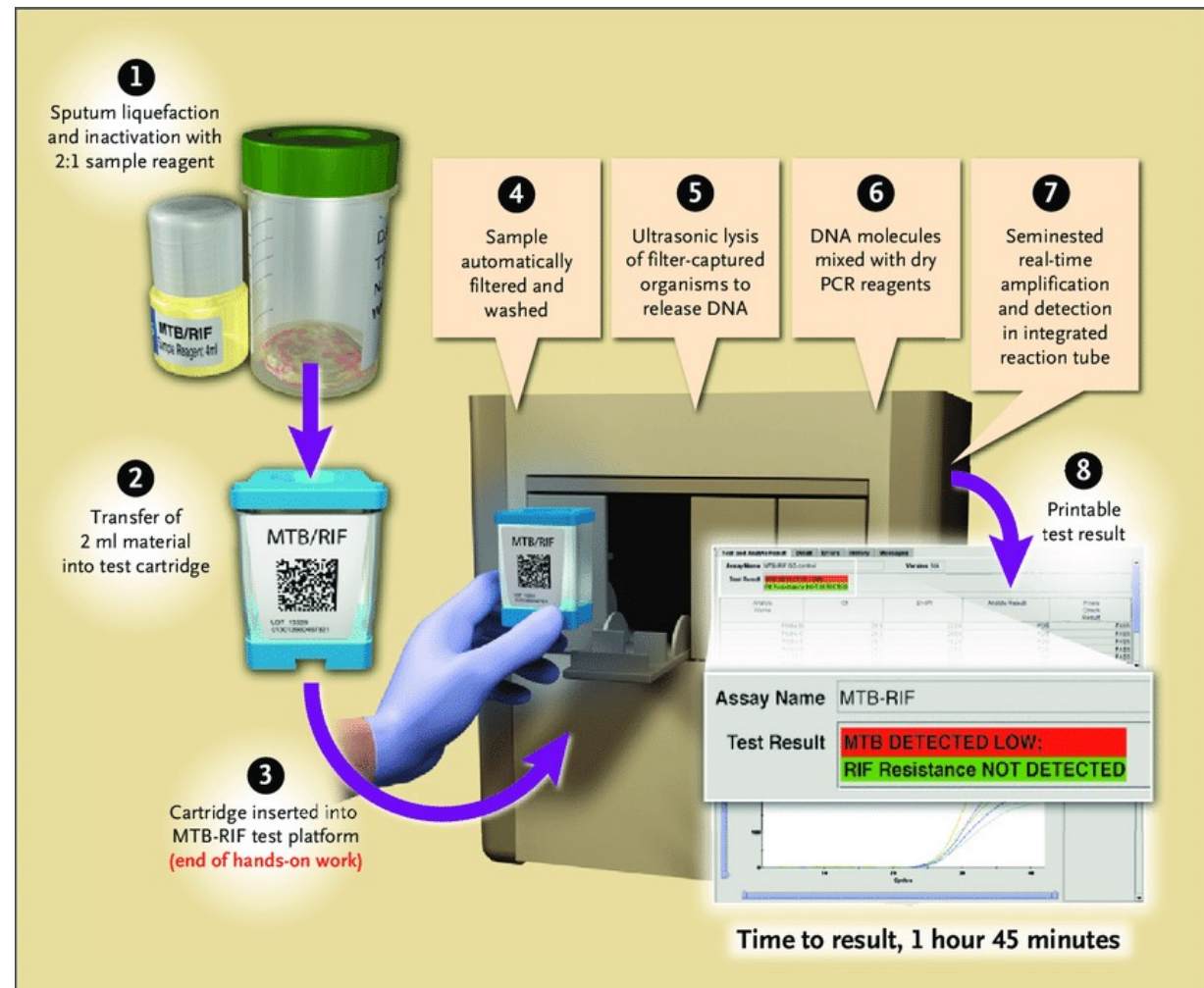
MTB DETECTED LOW;
Rif Resistance NOT DETECTED

Test and Analyte Result

Analyte Name	Ct	EndPt	Analyte Result	Probe Check Result
Probe D	28.2	148.0	POS	PASS
Probe C	27.1	225.0	POS	PASS
Probe E	28.5	126.0	POS	PASS
Probe B	27.8	134.0	POS	PASS
SPC	24.5	290.0	NA	PASS
Probe A	26.6	154.0	POS	PASS
QC-1	0	0	NEG	PASS
QC-2	0	0	NEG	PASS

PRACTICAL

- Work in groups of 2
- Eyeball sample volume
- Add 2x volume sample reagent
- Mix well
- Incubate at RT for 15 min
- Mix again at 10 min mark
- Add 2 ml to cartridge
- Load into machine
- Follow software prompts
- Wait for assay to run
- Interpret results



Deeplex Myc-TB

Targeted Next-Generation Sequencing



tNGS & WGS

Two NGS approaches for DR-TB control

tNGS

Focus on genomic
regions of interest



WGS

Comprehensive analysis
of the entire genome



tNGS & WGS

Two NGS approaches for characterisation of DR-TB

tNGS

Focus on genomic
regions of interest



WGS

Comprehensive analysis
of the entire genome



tNGS & WGS: Head to head

Characteristic	tNGS - <i>Deeplex® Myc-TB</i>	WGS
Input sample	Sputum	Culture
bp investigated	10,000 - 13000 bp	~ 4 million bp
Drug resistance	12 – 15 drugs fixed	23 drugs flexible
Limit of variant detection	1-3%	1-3% if 1000x depth
Phylogenetic Typing	Moderate resolution (SNPs/spoligotyping)	High resolution
Transmission Inference	Limited	Accurate

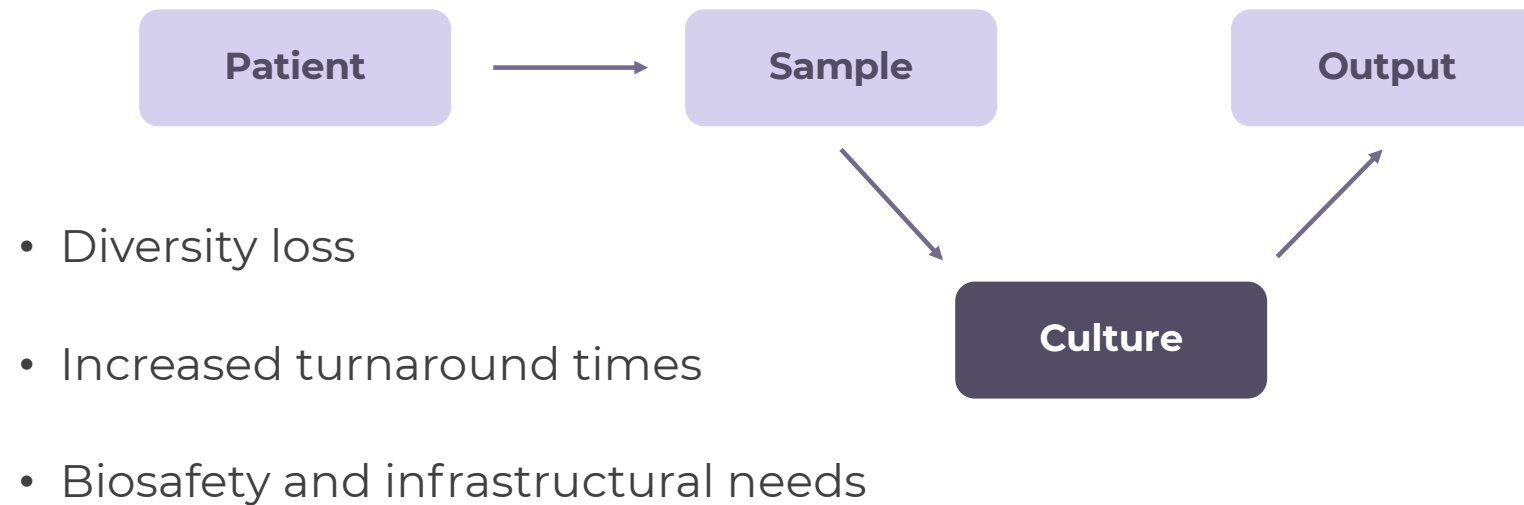
Implications of a culture step

tNGS - *Deeplex*® *Myc-TB*



Implications of a culture step

WGS



A comprehensive approach: Drug resistance

RIF	INH	EMB	PZA	BDQ	LZD	MFV	LFX	STM	AMK	KAN	CAP	CFZ	ETO
<i>rpoB</i>	<i>inhA</i>	<i>embB</i>	<i>pncA</i>	<i>mmpR5</i>	<i>rplC</i>	<i>gyrA</i>	<i>gyrA</i>	<i>rrs</i>	<i>rrs</i>	<i>rrs</i>	<i>rrs</i>	<i>mmpR5</i>	<i>inhA</i>
	<i>ahpA</i>				<i>rrl</i>	<i>gyrB</i>	<i>gyrB</i>	<i>rspL</i>		<i>eis</i>	<i>tlyA</i>		<i>ethA</i>
	<i>katG</i>							<i>gid</i>					

- 13-15 drugs
- 18 MTB drug-resistance associated targets



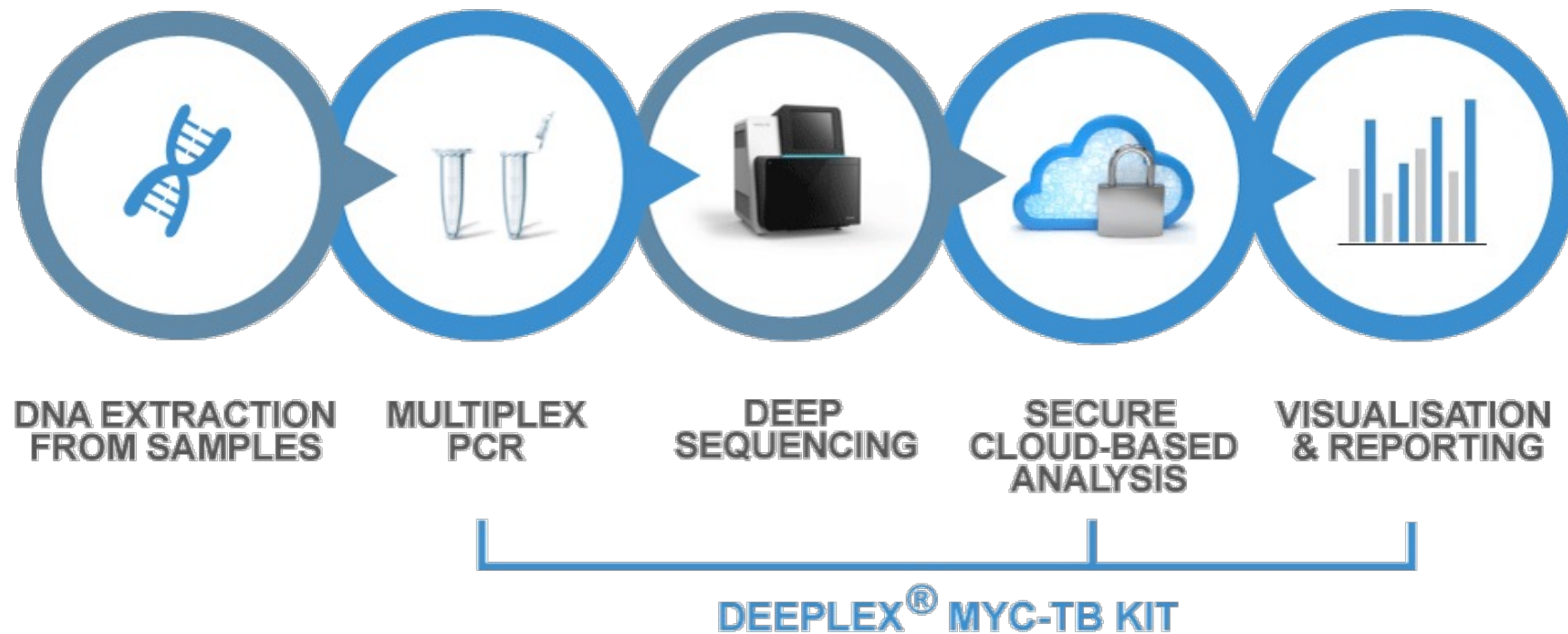
A comprehensive approach: Drug resistance

RIF	INH	EMB	PZA	BDQ	LZD	MFX	LFX	STM	AMK	KAN	CAP	CFZ	ETO
<i>rpoB</i>	<i>inhA</i>	<i>embB</i>	<i>pncA</i>	<i>mmpR5</i>	<i>rplC</i>	<i>gyrA</i>	<i>gyrA</i>	<i>rrs</i>	<i>rrs</i>	<i>rrs</i>	<i>rrs</i>	<i>mmpR5</i>	<i>inhA</i>
<i>rpoA</i>	<i>ahpA</i>	<i>embA</i>	<i>clpC1</i>	<i>mmpS5</i>	<i>rrl</i>	<i>gyrB</i>	<i>gyrB</i>	<i>rspL</i>	<i>whiB7</i>	<i>eis</i>	<i>tlyA</i>	<i>mmpS5</i>	<i>ethA</i>
<i>rpoC</i>	<i>katG</i>	<i>embC</i>	<i>panD</i>	<i>mmpL5</i>				<i>gid</i>	<i>eis</i>	<i>whiB7</i>	<i>whiB6</i>	<i>mmpL5</i>	<i>ethR</i>
<i>Rv2752c</i>	<i>mshA</i>	<i>embR</i>	<i>Rv1258c</i>	<i>atpE</i>				<i>whiB7</i>	<i>whiB6</i>		<i>ccsA</i>	<i>atpE</i>	<i>mshA</i>
	<i>Rv1258c</i>	<i>ubiA</i>	<i>PPE35</i>	<i>pepQ</i>				<i>Rv1258c</i>	<i>ccsA</i>		<i>fprA</i>	<i>pepQ</i>	<i>Rv3083</i>
	<i>ndh</i>		<i>Rv3236c</i>	<i>Rv1979c</i>				<i>whiB6</i>	<i>fprA</i>		<i>aftB</i>	<i>Rv1979c</i>	<i>ndh</i>
	<i>Rv2752c</i>								<i>aftB</i>				
DLM													
<i>fdg1</i>													
<i>ddn</i>													
<i>fbiA</i>													
<i>fbiB</i>													
<i>fbiC</i>													
<i>fbiD</i>													



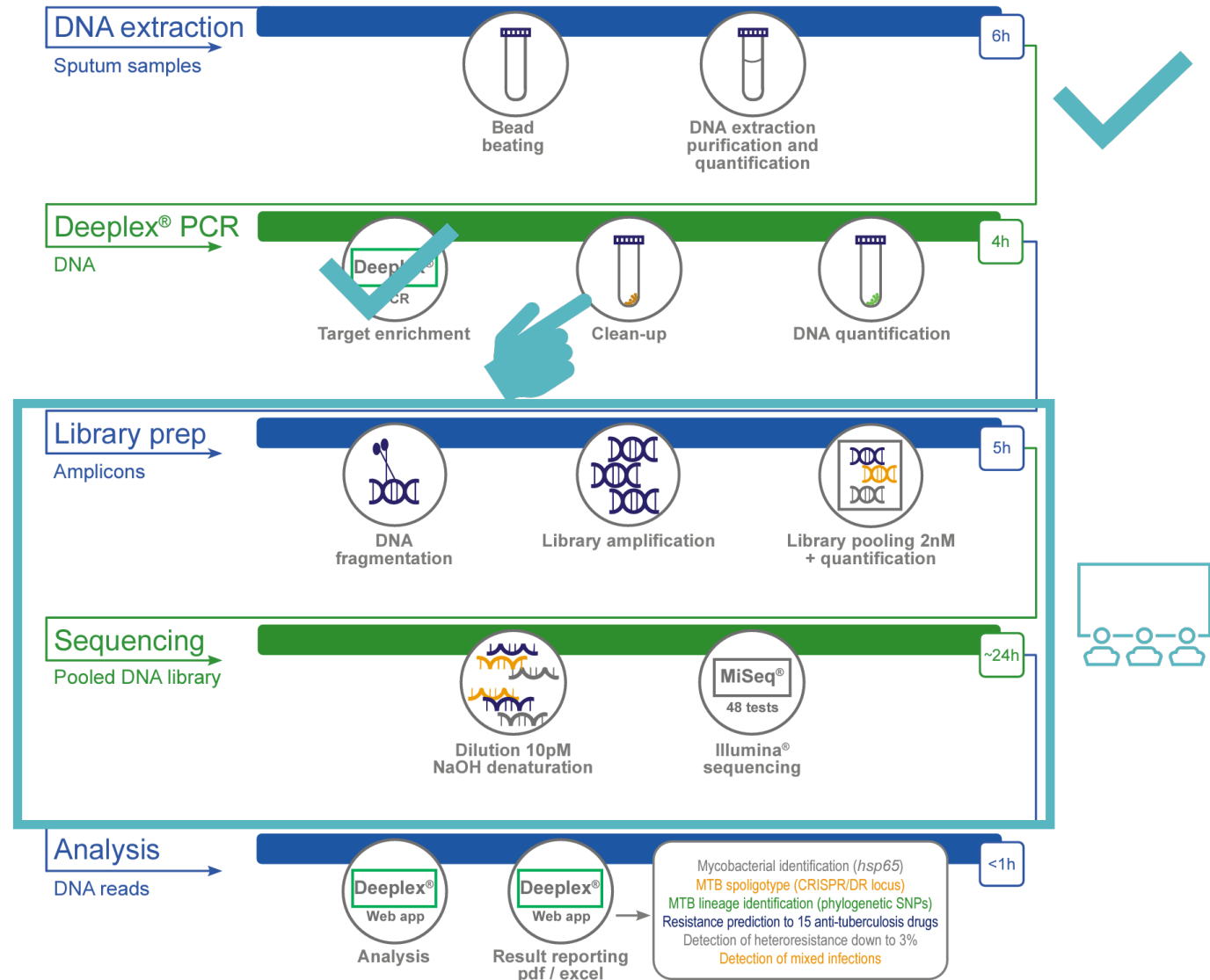
World Health Organization

PRACTICAL – Deeplex Myc-TB



PRACTICAL

Deeplex® Myc-TB workflow



Oxford Nanopore Technologies

Whole genome sequencing

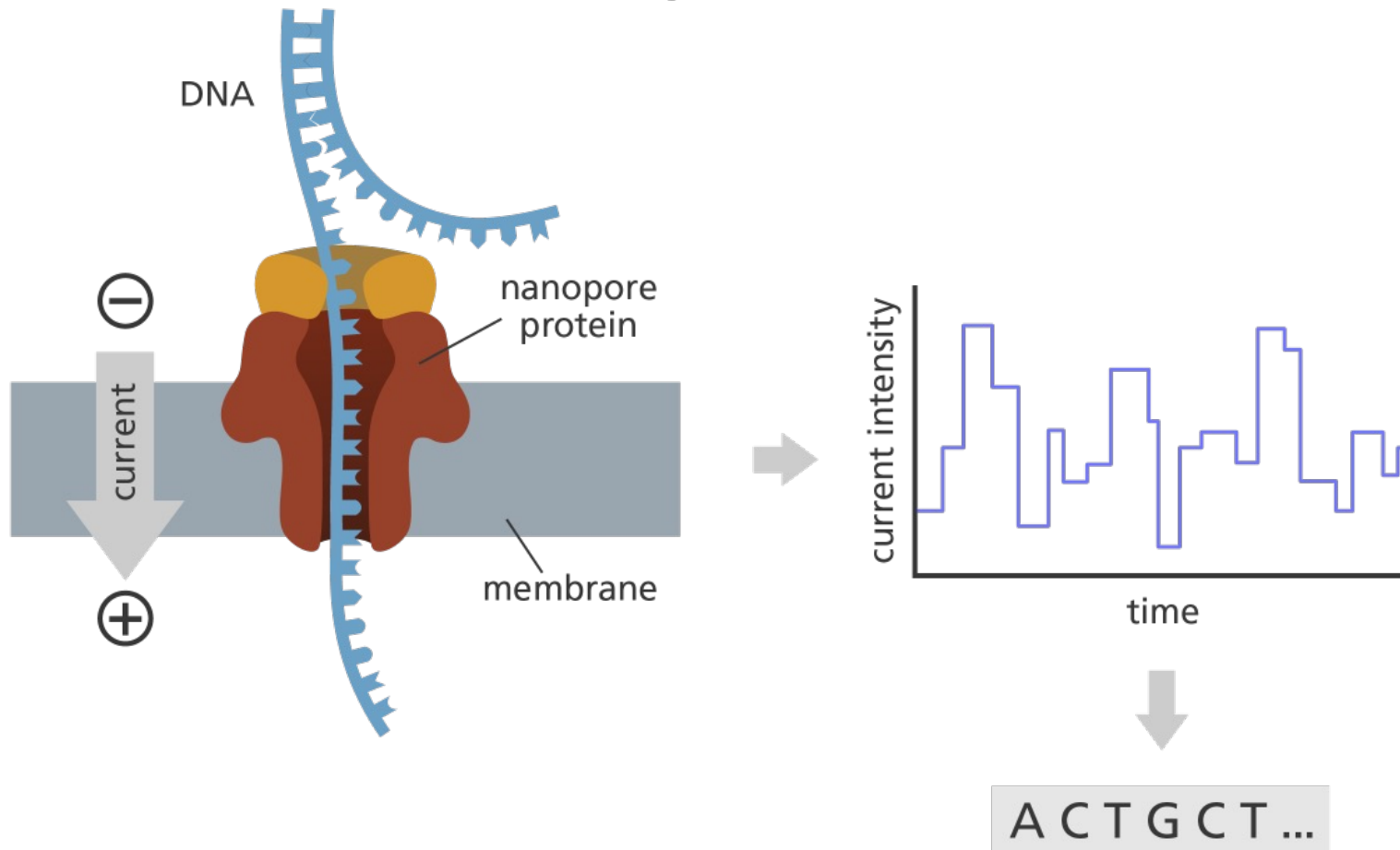




**Please make an account on Galaxy before
Thursday afternoon's session**

<https://usegalaxy.org/>

How ONT sequencing works - recap



<https://www.yourgenome.org/facts/what-is-oxford-nanopore-technology-ont-sequencing/>²⁴

Overview of ONT sequencing approaches

Investigations

Structural variation	SNVs and phasing
Gene expression	Identification
Splice variation	Assembly
Fusion transcripts	Epigenetics
Single cell	Chromatin conformation

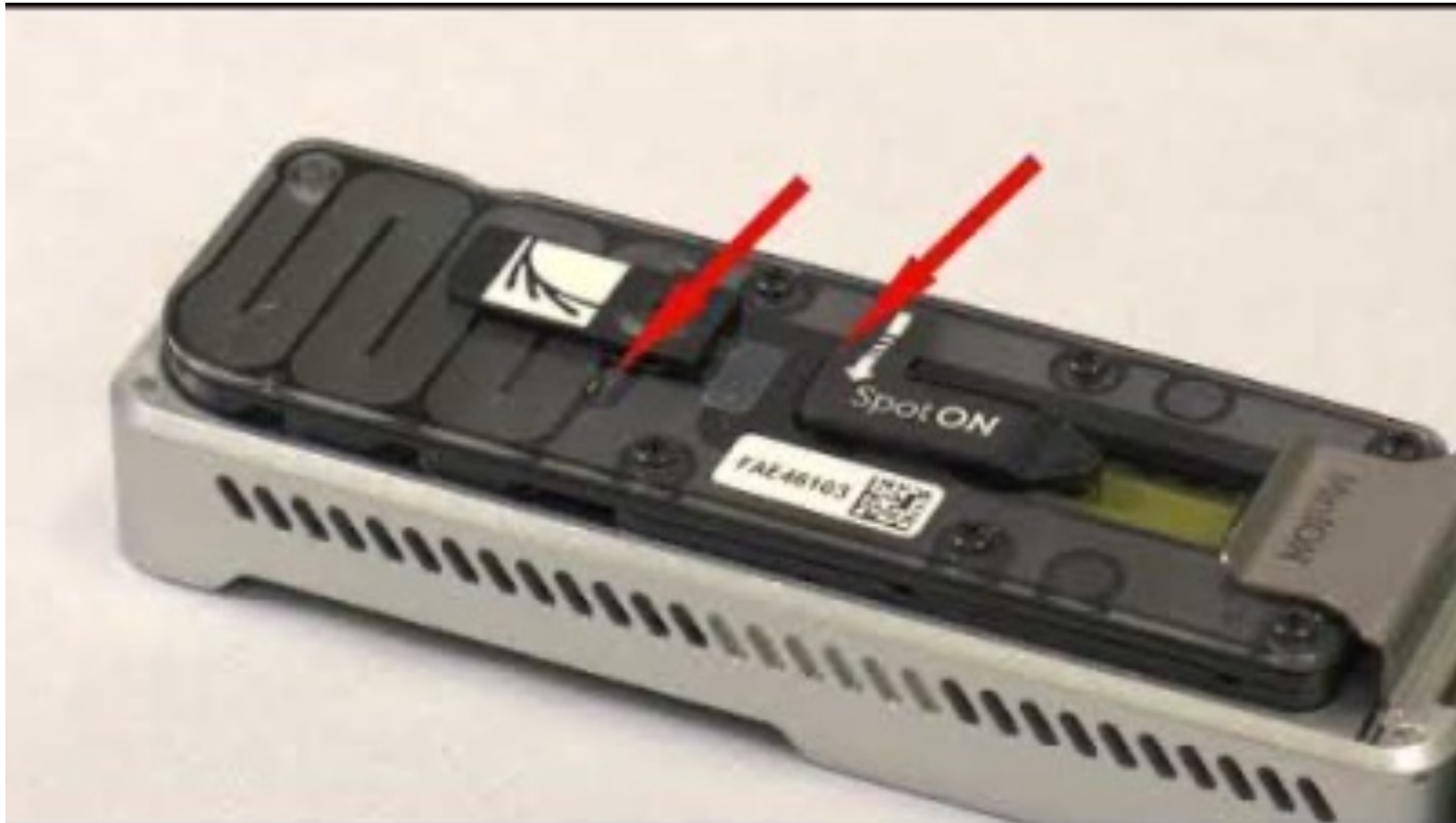
Techniques

Whole genome
Targeted
Whole transcriptome
Metagenomics
Short fragment mode

<https://nanoporetech.com/resource-centre>

<https://store.nanoporetech.com/>

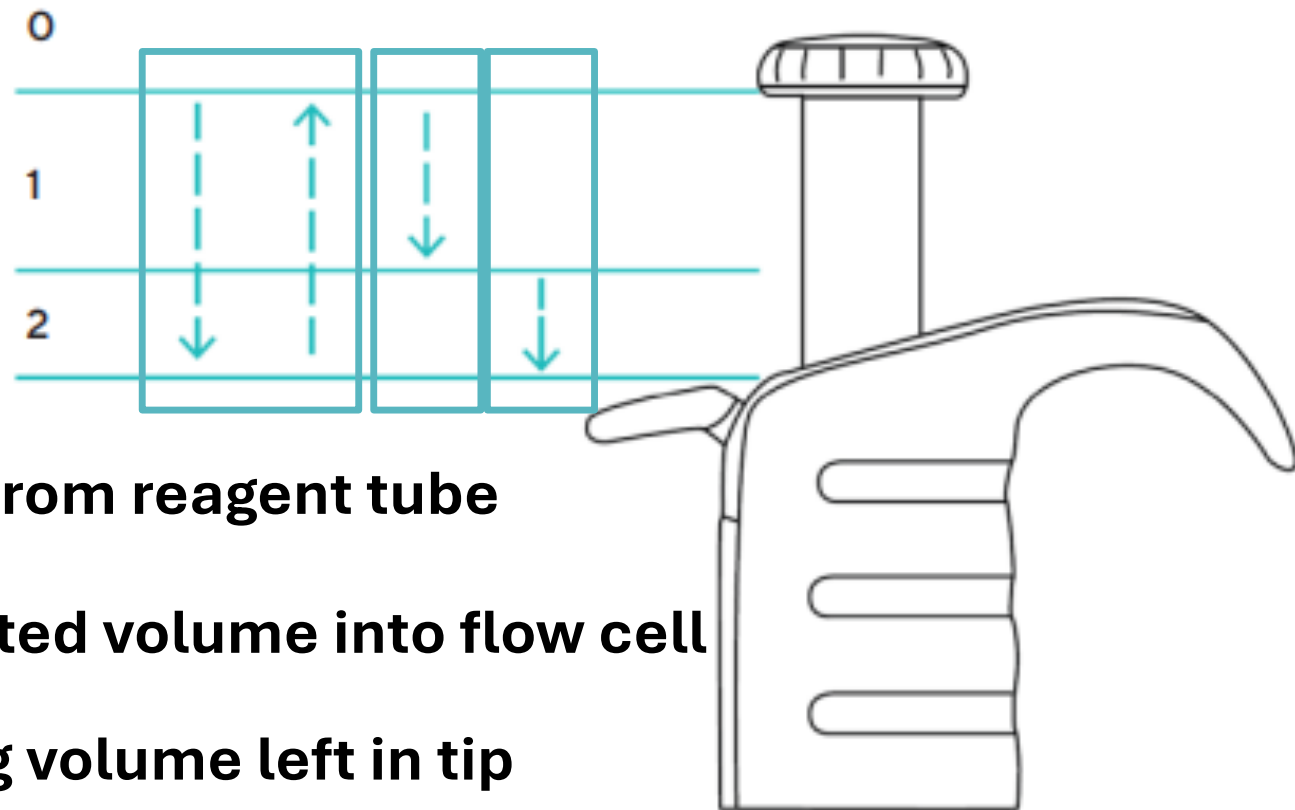
PRACTICAL – priming and loading the flow cell



Tips and tricks for loading a flow cell

- Removing liquid from the priming port to ensure that there are no bubbles in the sequencing channel
 - Insert the tip of a P1000
 - Slowly turn the dial to remove $\pm 200\mu\text{l}$ liquid
- Loading the flowcell: work quickly and accurately
 - Loading beads precipitate quickly
 - The order in which to open/close the ports matter
- Reverse pipetting
 - Use RP to add all reagents to the flow cell

Reverse pipetting



Collect reagent from reagent tube

Load only indicated volume into flow cell

**Empty remaining volume left in tip
in reagent tube**